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NEERAJ SINGH
DEVOPS

WHAT IS PRISMA CLOUD?

Prisma Cloud is Palo Alto Networks' Cloud-Native Application Protection Platform (CNAPP) that helps secure cloud-native applications from code to runtime across hybrid and multi-cloud environments.

PRISMA CLOUD OVERVIEW



CODE SECURITY

- SAST
- SCA
- IaC Security
- Secrets Detection

Find and fix security issues early in the SDLC.

CLOUD SECURITY

- CSPM
- CIEM
- Compliance
- Misconfigurations

Secure cloud infrastructure and identities.

RUNTIME SECURITY

- CWPP
- Container Security
- Kubernetes Security
- Serverless Security
- Runtime Protection

Protect workloads running in cloud environment.

KEY CAPABILITIES

CSPM (Cloud Security Posture)
Finds misconfigurations, security risks and compliance violations in AWS, Azure, GCP and other clouds.

CWPP (Cloud Workload Protection)
Protects VMs, containers, Kubernetes and serverless workloads.

CIEM (Cloud Identity Security)
Analyzes permissions and identifies excessive access.

IaC Security
Scans Terraform, CloudFormation and other infrastructure code.

Vulnerability Management (SCA)
Finds vulnerable open-source dependencies and packages.

Secrets Security
Detects exposed secrets and credentials in code, repositories and cloud.

WAAS (Web Application & API Security)
Protects web applications and APIs from common attacks (e.g., OWASP Top 10).

Runtime Security
Detects and helps prevent suspicious activity at runtime.

SUPPORTED ENVIRONMENTS

AWS · Azure · GCP | Kubernetes · Containers · VMs / Hosts · Serverless · Hybrid / Multi-Cloud

ONE PLATFORM → CODE-TO-CLOUD SECURITY

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PRISMA CLOUD + DEVSECOPS

Integrate Prisma Cloud in your CI/CD pipeline to find and fix security issues early before deployment.

DEVELOPER
Write code

↓

GITHUB
Push code

↓

CI/CD
Jenkins / GitHub Actions

↓

SAST
Scan source code for security issues

↓

SCA
Scan open source dependencies

↓

IaC SCAN
Scan Terraform / CloudFormation and other infrastructure code

↓

BUILD IMAGE
Build Docker image

↓

CONTAINER SCAN
Scan image for vulnerabilities and policy violations

↓

CONTAINER REGISTRY
Push secure image

↓

DEPLOY TO KUBERNETES
Run secure workloads

PRISMA CLOUD SECURITY CHECKS

SAST	Detects code vulnerabilities (e.g., OWASP Top 10)
SCA	Finds vulnerable open source dependencies
IaC SECURITY	Checks Terraform, CloudFormation, Kubernetes manifests for misconfigurations
CONTAINER SECURITY	Scans container images for CVEs and policy violations
POLICY ENFORCEMENT	Enforces organizational security policies and compliance

EXAMPLE: SECURITY GATE

Container Image Scan

↓

Critical Vulnerabilities found?

YES → **BLOCK DEPLOYMENT**
- Fail the pipeline
- Notify team
- Fix and rebuild

NO

ALLOW DEPLOYMENT
- Push to registry
- Deploy to Kubernetes

SHIFT SECURITY LEFT

- Find issues early
- Fix before deployment
- Reduce risk in production

BENEFITS

SECURE PIPELINE	COMPLIANCE	FASTER REMEDIATION	SAFE DEPLOYMENTS
Detect and prevent vulnerabilities early	Meet security and regulatory requirements	Fix issues in development stage	Reduce risks in production

FIND EARLY → FIX EARLY → DEPLOY SECURELY

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CSPM + CIEM in PRISMA CLOUD

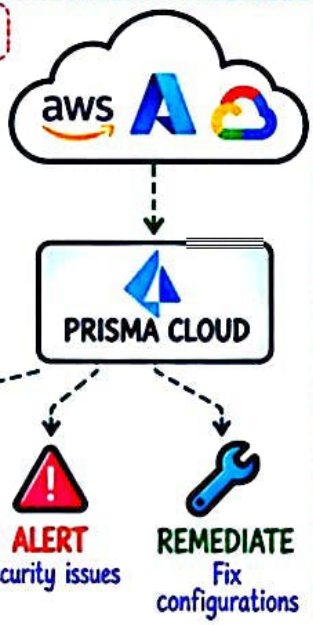
Secure your cloud environment by fixing misconfigurations and managing identities with least privilege access.

CSPM (CLOUD SECURITY POSTURE MANAGEMENT)

CSPM continuously scans your cloud environment for misconfigurations, security risks and compliance violations.

WHAT CSPM CHECKS ?

- Publicly accessible storage
- Open security group rules
- Encryption disabled
- Risky configurations
- Compliance violations (e.g., CIS, NIST, SOC2)
- Unmanaged cloud resources



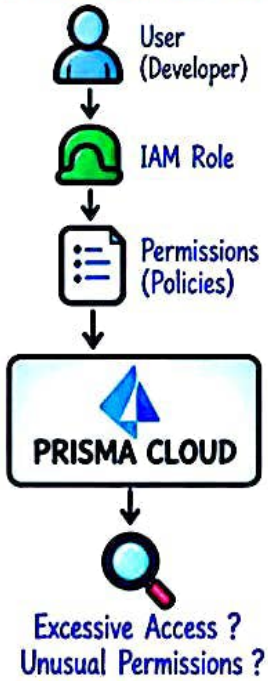
EXAMPLE FINDINGS

Cloud Resource	Risk	Impact
S3 Bucket (public)	Public access enabled	Data leak
Security Group	0.0.0.0/0 on port 22	Unauthorized access
RDS	Encryption disabled	Sensitive data exposure
IAM Role	Overly permissive policies	Privilege escalation

CIEM (CLOUD INFRASTRUCTURE ENTITLEMENT MANAGEMENT)

CIEM analyzes cloud identities, permissions and access paths to identify excessive access and helps enforce least privilege permissions.

HOW CIEM WORKS ?



CIEM IDENTIFIES

- Overly permissive roles
- Unused permissions
- Risky trust relationships
- Excessive access to sensitive resources
- Lateral movement paths
- Violation of least privilege

EXAMPLE



SUPPORTED CLOUD PROVIDERS

aws AWS | Azure | GCP | Hybrid / Multi-Cloud

SECURE CONFIGURATION + SECURE IDENTITY = STRONGER CLOUD SECURITY

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CONTAINER, KUBERNETES & RUNTIME SECURITY

Prisma Cloud protects your containerized and cloud workloads by scanning images, enforcing security policies and detecting threats at runtime.

CONTAINER SECURITY FLOW

Source Code
Scan for secrets, SAST, SCA

↓

Build Image
Scan image for vulnerabilities and policy violations

↓

Container Registry
Push secure image

↓

Kubernetes Deployment
Deploy to cluster

↓

Running Workloads
Monitor and protect pods, containers and hosts

↓

Runtime Protection
Detect suspicious behavior, threats and policy violations

EXAMPLE: CONTAINER IMAGE SCAN

App Image v1.0 → Prisma Cloud Image Scan →

Vulnerabilities Found
- 2 Critical
- 5 High
- 3 Medium

↓

Block Deployment
Based on security policy

WORKLOADS PROTECTED BY PRISMA CLOUD

Containers
Scan images and protect containers

Kubernetes
Secure clusters, pods and workloads

VMs / Hosts
Protect VMs and hosts in cloud/on-prem

Serverless
Secure serverless functions (e.g., AWS Lambda)

Web Applications & APIs
Detect and protect web applications and APIs from common attacks (e.g., OWASP Top 10).

RUNTIME THREAT DETECTION

Suspicious Process → **Detect** abnormal activity in real-time → **Alert** Generate alerts with context → **Investigate** View process, file, network details → **Respond** Take action (block, isolate)

BENEFITS

Prevent Threats	Detect and block threats at runtime
Enforce Policies	Ensure security and compliance
Reduce Risk	Find and fix vulnerabilities early
Improve Visibility	Full visibility across code, images, registries and running workloads

BUILD-TIME SECURITY + RUNTIME SECURITY = SECURE WORKLOADS

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PRISMA CLOUD QUICK REVISION

Key components, interview points and important takeaways for DevOps Engineers.

CSPM
(Cloud Security Posture)

- Find misconfigurations
- Detect security risks
- Ensure compliance

CWPP
(Workload Protection)

- Protect VMs, containers
- Detect vulnerabilities
- Prevent runtime threats

CIEM
(Identity Security)

- Analyze permissions
- Find excessive access
- Enforce least privilege

IaC SECURITY
(Infrastructure as Code)

- Scan Terraform, CloudFormation
- Detect misconfigurations
- Prevent insecure deployments

VULNERABILITY MANAGEMENT

- Find vulnerable packages
- Prioritize critical risks
- Track and remediate

RUNTIME SECURITY

- Detect suspicious activity
- Protect running workloads
- Respond to threats

INTERVIEW QUICK CHECK

1. **Q: What is Prisma Cloud?**
   A: A CNAPP platform for securing cloud-native applications from code to runtime.
2. **Q: Why integrate Prisma Cloud with CI/CD?**
   A: To identify and fix security issues early and enforce security policies before deployment.
3. **Q: What is CSPM?**
   A: Detects misconfigurations and security risks in cloud resources.
4. **Q: What is CWPP?**
   A: Protects workloads like VMs, containers and Kubernetes.
5. **Q: What is CIEM?**
   A: Analyzes identities, permissions and access paths to enforce least privilege access.

SUPPORTED ENVIRONMENTS

AWS | Azure | GCP

Kubernetes | Containers | VMs / Hosts | Serverless

KEY TAKEAWAYS

1. One platform for code-to-cloud security
2. Detect and fix issues early
3. Protect cloud workloads at runtime
4. Enforce security policies and compliance
5. Reduce risk and improve visibility

PRISMA CLOUD

CODE → SCAN → BUILD → DEPLOY → PROTECT → MONITOR

SECURITY FROM CODE TO CLOUD TO RUNTIME